

Sep 2021 – Present	Ph.D. Student in the Department of Atmospheric and Oceanic Sciences University of California, Los Angeles: <i>Thesis Proposal</i> : “Constrain Oceanic Mesoscale Energy Dissipation”; <i>Advisors</i> : Dr. Andrew Stewart and Dr. James McWilliams (Dec. 2026 expected degree completion, Dec. 2024 candidacy, Dec. 2023 M.S.)
Sep 2018 – Dec 2019	M.S. in the Graduate Institute of Hydrological and Oceanic Sciences, National Central University, Taiwan: <i>Thesis</i> : “A UAV-RTK-Lidar system measurements of wave energy dissipation over a sandy beach and an algal-reef area”; <i>Advisor</i> : Dr. Zhi-Cheng Huang
Sep 2014 – Jun 2018	B.S. in the Department of Earth Sciences, National Central University, Taiwan

PEER-REVIEWED PUBLICATIONS

3. C. Yeh, A. L. Stewart, J. C. McWilliams, 2026: Satellite- and Argo-Constrained Global Estimates of Near-Seafloor Mesoscale Eddy Kinetic Energy and Bottom Drag Dissipation Rates. *In prep for GRL*
2. C. Yeh, A. L. Stewart, J. C. McWilliams, 2026: Regimes of oceanic mesoscale energy dissipation at western boundaries. *accepted by Journal of Physical Oceanography*, DOI: 10.1175/JPO-D-25-0059.1
1. Huang, Z., C. Yeh, K. Tseng, and W. Hsu, 2018: A UAV-RTK Lidar system for wave and tide measurements in coastal zones. *Journal of Atmospheric and Oceanic Technology*, 35, 1557–1570, 2018, 10.1175/JTECH-D-17-0199.1.

PROFESSIONAL SKILLS AND PUBLISHED TOOLS

MATLAB, Geophysical Fluid Dynamics (Rossby wave dynamics, shallow water theory, quasi-geostrophic theory, isopycnal coordinate, energy budget), MATLAB Parallel Computing, Global current-meter dataset GMACMD (Scott et al. 2010, 2011) analysis, rotating tank, Unmanned Aerial Vehicle techniques, Fourier analysis, surface waves theory, Empirical Orthogonal Functions analyses, EOF_vertical (https://github.com/sunnyyeh3725/EOF_vertical.git)

FELLOWSHIP AND COMPUTATIONAL SUPPORT

Apr 2025 – Jun 2025	J Yang Fellowship, \$14,000, Asia Pacific Center, University of California, Los Angeles
Sep 2021 – Jun 2024	Government Scholarship to Study Abroad, \$147,000, Ministry of Education, Taiwan
May 2025 – May 2026	“EES250071: Using Machine Learning and Satellite Measurement to Estimate Global Bottom Drag Dissipation”, 200,000 ACCESS credits, EXPLORE Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support (ACCESS), P.I. Cheng Yang Yeh

HONORS AND AWARDS

- *Outstanding Student Presentation Award*, California Geophysical Fluid Dynamics (CalGFD) meeting, 2023, Scripps, La Jolla, CA
- *Young Scientist Award*, The 9th Chinese-German Joint Symposium on Hydraulic and Ocean Engineering, 2018, National Cheng Kung University, Taiwan
- *Youth Forum 3rd Award*, Taiwan Ocean Science Conference, 2019, Taipei, Taiwan
- *Poster 2nd Award*, Taiwan Ocean Science Conference, Taipei, Taiwan
- *Best Student Award*, National Central University, 2019 Fall semester (60 students awarded per semester in the university)
- *Academic Award Scholarship*, Department of Earth Sciences, National Central University

EMPLOYMENT

Jan 2022 – Present	Graduate Student Researcher: (Appointment during quarters not assigned to teaching) <i>PI: Andrew Stewart and James McWilliams</i> Teaching Assistant: (Fall 2022, 2023, 2024, 2025, Class: AOS103) <i>PI: Andrew Stewart</i> Appointments are in Department of Atmospheric and Oceanic Sciences, UCLA
Sep 2020 – Jun 2021	Research Assistant, <i>COAST group PI: Zhi-Cheng Huang</i> , Graduate Institute of Hydrological and Oceanic Sciences, National Central University
Dec 2019 – Mar 2020	Research Assistant, <i>COAST group PI: Zhi-Cheng Huang</i> , National Central University
Jan 2016 – Feb 2016	Full-time Internship, <i>Sinotech Engineering Consultants, Ltd.</i> , Taipei, Taiwan

SEMINAR TALKS

Formal:	Institute of Oceanography, National Taiwan University, “ <i>An idealized framework and global data approach to constrain oceanic mesoscale energy dissipation</i> ”, January 2026 Department of Atmospheric Science, National Central University, “ <i>An idealized framework and global data approach to constrain oceanic mesoscale energy dissipation</i> ”, January 2026 Department of Ocean Science, The Hong Kong University of Science and Technology, “ <i>Regimes of oceanic mesoscale energy dissipation at western boundaries</i> ”, March 2024
Informal:	ODSL group, PI: Jinbo Wang, Department of Oceanography, Texas A&M University, “ <i>Constraining Oceanic Mesoscale Energy Dissipation</i> ”, July 2025 Institute of Oceanography, National Taiwan University, “ <i>Regimes of oceanic mesoscale energy dissipation at western boundaries</i> ”, Mar 2024

TEACHING

The Physical Oceanography TA is a weekly discussion class in the quarter system at the University of California, Los Angeles. The Earth Hydrological Cycle TA is a weekly quiz TA sessions in the semester system at National Central University, Taiwan. The AOS Summer Lecture is a week of oceanic topic discussion for new incoming Ph.D. students. The Class Lecture is a 75-minute UCLA upper-division lecture. I designed and taught the UAV and Pix4D Image Processing class, which is a 1-credit lecture in the semester system at the National Central University.

Teaching Assistant:	Physical Oceanography (Upper-division)	Fall 2022, Fall 2023, Fall 2024, Fall 2025
	Earth Hydrological Cycle (Upper-division)	Spring 2018, Spring 2019
	AOS Summer Lecture	Summer 2022
Class Lecture:	Physical Oceanography (Upper-division)	Fall 2023 (Eddies and Turbulence), Fall 2024 and 2025 (Internal Waves)
Instructor:	UAV and Pix4D Image Processing (Under-graduate)	Fall 2018, Fall 2019, Fall 2020, Spring 2021

ACADEMIC SERVICE

Leadership activities:	<i>Chair of the Marine Center Graduate Student and Postdoctoral Network</i> , Marine Center, University of California, Los Angeles, Sep 2023 – present
Committee:	<i>270 Seminar Committee</i> , Department of Atmospheric and Oceanic Sciences, University of California, Los Angeles, Sep 2023 – present
Conference services:	<i>CalGFD Conference Organizing Team</i> , 2024, 2025
Public Outreach:	<i>Research Night Panel speaker</i> , Department of Atmospheric and Oceanic Sciences, University of California, Los Angeles, Oct 2023 <i>Rotating tank demonstration</i> , Explore Your Universe Science Fair, UCLA, Nov 2023

MENTORSHIP

Research:	Aisha Mardini, “ <i>Using Machine Learning and Satellite Measurement to Estimate Global Bottom Drag Dissipation</i> ”, May 2025 – present
	Tanner Shinkle, “ <i>Observing Surface Submesoscale Flows with Unmanned Aerial Drones</i> ”, Oct 2023 – Jun 2024
Academic:	Yuzuna Kudo, Sep 2024 – present
	Elena Dworak, Sep 2023 – Jun 2024
	Athena Bamrick, Sep 2021 – Jun 2023

WORKSHOPS

1. “*NASA Ocean AI Workshop*”, Caltech, Pasadena, May 19th – 21st 2025

SELECTED CONFERENCE ABSTRACTS

11. “*Estimating global ocean bottom drag dissipation via Empirical Orthogonal Function analysis*”, Ocean Science Meeting, Glasgow, Scotland, Feb 23rd – 27th 2026
10. “*Estimating global ocean bottom drag dissipation via Empirical Orthogonal Function analysis*”, Climate Processing Team Annual Meeting, New York University (online presentation due to UCLA travel ban), Aug 13th – 15th 2025
9. “*Constraining Global Oceanic Mesoscale Energy Dissipation: The Role of the Ocean Bottom*”, UCLA Marine Center Annual Meeting, Jun 12th 2025
8. “*Regimes of oceanic mesoscale energy dissipation at western boundaries*”, Climate Processing Team Annual Meeting, Brown University, Providence, RI, Aug 14th – 16th 2024
7. “*Regimes of oceanic mesoscale energy dissipation at western boundaries*”, Ocean Science Meeting, New Orleans, La, Aug 18th – 23rd 2024
6. “*Regimes of oceanic mesoscale energy dissipation at western boundaries*”, CalGFD 2023, Scripps, La Jolla, CA
5. “*Regimes of oceanic mesoscale energy dissipation at western boundaries*”, UCLA Marine Center Annual Meeting, May 22th 2023
4. “*A UAV-RTK-Lidar system measurements of wave energy dissipation over a sandy beach and an algal-reef area*”, Coastal Engineering No. 36v, doi.org/10.9753/icce.v36v.waves.34, <https://youtu.be/fBLbsBuK4AA>, 2020
3. “*Understanding the variations of sand coverage on a reef system in the intertidal zone of Taoyuan coast using airborne imagery techniques*”, AGU Fall Meeting, 2019
2. “*A UAV-RTK-Lidar system measurements of wave energy dissipation over a sandy beach and an algal-reef area*”, EGU meeting, EGU2019-12158, 2019
1. “*A UAV-RTK-Lidar System for Wave and Tide Measurements in Coastal Zones*”, The 9th Chinese-German Joint Symposium on Hydraulic and Ocean Engineering, National Cheng Kung University, Taiwan, 2018